

EXAM 7 – SPRING 2015

8. (3.5 points)

An insurer writes the following lines of business and has derived the associated coefficients of variation (CoV) for claim liabilities for the purpose of determining a risk margin.

	Claim Liability <u>Distribution</u>	Internal Systemic <u>Risk CoV</u>	External Systemic <u>Risk CoV</u>	Independent Risk <u>CoV</u>
<u>Line of Business</u>				
Personal Auto	60%	5%	4%	5%
Commercial Auto	35%	7%	5%	5%
Commercial Umbrella	5%	15%	10%	7%

- The insurer will only write umbrella over its own commercial auto policies.
- The insurer used a balanced scorecard approach for internal systemic risk.

a. (1.5 points)

Describe two sources of internal systemic risk. Briefly explain why each one might lead to a higher CoV selection for umbrella than for the other two lines of business.

b. (0.75 point)

Identify a source of external systemic risk with a relatively *high* correlation between two of the above lines of business. Name the two lines and provide a brief reason for the high correlation.

c. (0.75 point)

Identify a source of external systemic risk with a relatively *low* correlation between two of the above lines of business. Name the two lines and provide a brief reason for the low correlation.

d. (0.5 point)

Assuming independence, calculate the independent risk CoV for the overall portfolio.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 8	
TOTAL POINT VALUE: 3.5	LEARNING OBJECTIVE: A5: Describe the various sources of risk and uncertainty that are associated with the determination of reserves. Calculate risk margins that consider these sources of risk and uncertainty.
SAMPLE ANSWERS (BY PART, AS APPLICABLE)	
Part a: 1.5 points	
<i>Sample Answer 1</i> Specification Error – the risk that underlying process is too complex to select a model that fully explained the insurance process. Umbrella claims are inherently more variable due to their high attachment and longer tail. Parameter Selection Error – the risk that the model is unable to measure accurately the predictors in claim cost or trends in those predictors. Certain trends, like severity trend, have larger impacts on excess layers – for umbrella this will create more importance of getting those factors right. <i>Sample Answer 2</i> Specification Error – risk associated with the fact you can't develop a perfect model for insurance because it is too complex. Umbrella is a much less homogeneous line than PA and CA and also a low frequency high severity line so we anticipate higher volatility and therefore higher CoV Data Error – risk associated with errors in the data, or lack of understanding of the data, or unreliable data. Umbrella is a much more nuanced line than PA or CA, with fewer industry statistics, so fewer benchmarks and in general there is less industry expertise, so the chances for data to be unavailable or for expertise of understanding the data to be low is much greater.	
Part b: 0.75 point	
<i>Sample Answers</i> • Catastrophe risk would affect both personal and commercial auto because a catastrophe would affect an entire area and if both personal and commercial auto are in that area then you will see large losses in both lines. • Economic risk such as inflation, fuel prices – personal auto and commercial auto are both subject to the same inflation in terms of the replacement cost of vehicles & vehicle parts. If costs of the replacement parts increased in one line, it will increase in the other as well. • Recovery risk is highly correlated between PA & CA because it is risk associated from recoveries, like from damaged cars. • Claims management changes b/w PA & CA b/c both would fall under the same chief claims officer & changes in claims handling would likely affect both. • Regulation/Political Risk. Both personal and commercial auto will be subject to the same legal shifts and regulatory requirements. Because both offer the same general types of coverages, changes to minimum BI policy limits (For example) will impact both lines.	
Part c: 0.75 point	

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Sample Answers

- Claim Management Process Risk – Changes in settlement, reporting, finalization of claims. PA and CU unlikely to be handled by the same claims department.
- Legal (Political/Legislative Risk) – personal auto is much more regulated than commercial auto so it is unlikely that any regulatory or legal changes would impact both.
- Political and legal risk between personal auto and commercial umbrella is likely lower because most political attention regarding legal insurance required, rating, etc. is in regards to personal insurance. There is not as much regulation of umbrella coverage.
- Latent claim can have a low correlation between personal auto and umbrella as personal auto is short tail and is not likely that a latent claim, say asbestos, can affect personal auto and commercial umbrella at the same time as their cause of loss will be very different.
- Event risk: personal auto and commercial umbrella are unlikely to be impacted by any one event. For example, an event would cause damage to PA but not CU as CU is just liability.
- Recovery risk – recovery from PA are mainly subro and salvage from other insurers. Recovery from CU are mainly by reinsurance. Therefore, the recovery risk of PA has low correlation with that of CU.
- Personal auto & commercial auto would have low correlation for expense risk. The two have different claims units – one entry-level and systemized, the other highly skilled and expensive. Further, umbrella would use attorney's more frequently and claims would volatility would lead to expense volatility personal wouldn't have.
- Claims management process between personal auto and commercial auto are low because the insurers likely to have separate claims staff handle comm and personal claims. Change in one is unlikely to be implemented in the other – different practices.

Part d: 0.5 point

Sample Answer 1

$$= \text{SQRT} [(.60 * 5\%)^2 + (.35 * 5\%)^2 + (.05 * 7\%)^2]$$

$$= .035$$

Sample Answer 2

$$= \sqrt{(.60 * .05)^2 + (.35 * .05)^2 + (.05 * .07)^2}$$

$$= 0.0349$$

EXAMINERS' REPORT

Overall, candidates performed well on this question. Detailed commentary provided by part below.

Part a

Candidates were expected to be able to describe two of the main sources of internal systemic risk, along with a possible reason for a higher umbrella CoV for each of the two sources.

On the first part (describing sources of internal systemic risk), candidates could earn full credit either for identifying the risk and providing a brief description, or for giving a more robust description, in which case an identification was not necessary. For the identification, candidates were given credit for writing either “model” or “specification” error.

For data error (an internal systemic risk source), defining data error as solely the risk of having little data, without any further explanation, did not earn credit.

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For explaining the higher umbrella CoV, several reasons were acceptable. Generally anything that demonstrated an understanding of the complexity/nature of the umbrella line was given credit.

While candidates generally did well, some common errors include candidates identifying, but not briefly describing the two sources of internal systemic risk and in general, insufficient explanations.

Part b

Candidates were asked to give a source of external systemic risk and correctly identify two lines that have high correlation. They were also expected to explain why the two lines have high correlation for the risk given.

For the explanation portion of this question, several reasons were acceptable. Demonstrating an understanding of why two lines would have high correlation earned credit.

The vast majority of candidates received full credit for this part.

The most common error was to give a flawed or insufficient explanation for why the two lines were correlated.

Part c

Candidates were expected to give a source of external systemic risk and correctly identify two lines that have low correlation. They were also expected to give a proper explanation of why the two lines have low correlation for the risk given.

For the explanation portion of this question, several reasons were acceptable. Demonstrating an understanding of why two lines would have low correlation was given credit.

Generally candidates did well, but some struggled to give a good explanation.

One common error was to give a flawed explanation of why two lines saw low correlation. Another common error was for the candidate to identify two lines that should properly be high correlation for the risk given combined with an example of an uncorrelated event for the two lines that ignores a more global perspective of correlations. A prevalent example of this error was to provide one economic scenario that might not affect both lines while ignoring the fact that all lines are affected by inflation. A variant of this example was to confuse difference in magnitude with the impact for low correlation.

Part d

Candidates were expected to calculate the independent CoV of the three lines of business, assuming independence.

Generally candidates did well on this question. Common errors included not applying the weights for each line of business, using the wrong set of CoVs, and minor calculation errors.